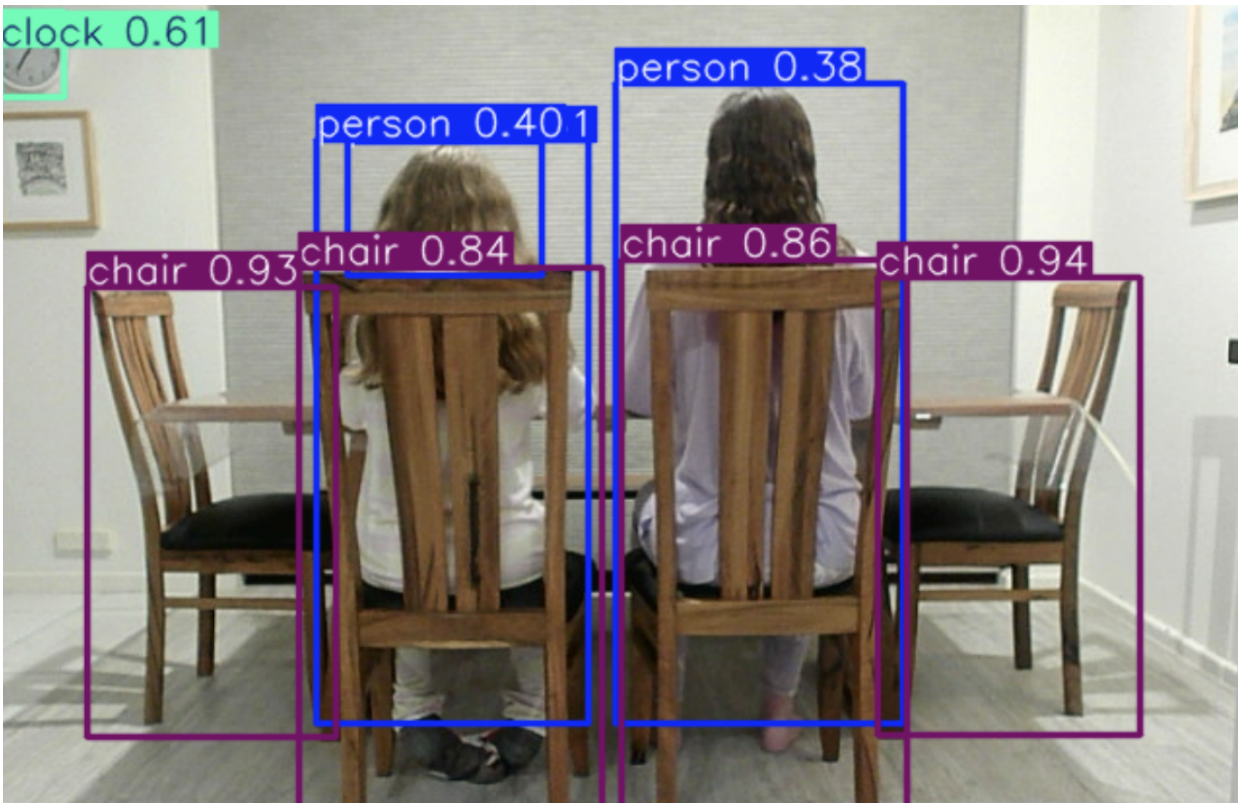


Figure 2.4 An example of using a multi object-detection computer vision model over a video stream using the YOLO v8 model to determine room occupancy. The model draws a bounding box around each object it detects in the stream, and adds a label to the box with a confidence score, which is a combined percentage representing the model's certainty that a detected object belongs to a specific class and that the bounding box contains an object.

In my home digital twin, I can make use of edge object detection in a video stream to detect occupancy levels in the home, and use this data to make decisions about energy or water use.

While the use of video data to capture physical systems for digital representation requires careful consideration of architecture, privacy, and data management, the resulting value extends far beyond simple visualization, creating digital twins that truly mirror the living, changing nature of physical systems. As computational capabilities and computer vision algorithms continue to advance, video will increasingly serve as a foundational data source for digital twins across industries, enabling levels of fidelity and responsiveness previously unattainable.